

3. Write a program to find the n^{th} fibonacci number using recursion

→ I/P

```
# include <stdio.h>
```

```
int fibonacci(int n){
```

```
    if (n==0)
```

```
        return 0;
```

```
    else if (n==1)
```

```
        return 1;
```

```
    else
```

```
        return fibonacci(n-1) + fibonacci(n-2);
```

```
}
```

```
int main(){
```

```
    int n;
```

```
    printf("Enter n:");
```

```
    scanf("%d", &n);
```

```
    printf("Fibonacci number = %d", fibonacci(n));
```

```
    return 0;
```

```
}
```

O/P :-

Enter n = 11

Fibonacci number = 89

main.c



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Run

Output

Clear

```
1 #include <stdio.h>
2
3 unsigned long long fib(int n)
4 {
5     if (n <= 1)
6         return n;
7     return fib(n - 1) + fib(n - 2);
8 }
9
10 int main()
11 {
12     int num;
13
14     printf("Enter no: ");
15     scanf("%d", &num);
16
17     printf("%dth Fibonacci: %llu\n", num, fib(num));
18
19     return 0;
20 }
```

Enter no: 11
11th Fibonacci: 89

=== Code Execution Successful ===